

Xie Jiayu Dias

Data Engineer & AI Researcher

Data/AI engineer and researcher with a background in Electronic Engineering and a Master's in Computer Vision from Marquette University. Experienced in building end-to-end data pipelines, ML models, and integration systems across industry (ExxonMobil, Lactec ¹) and academia. Passionate about mentoring and STEM outreach, actively engaged in step-out / volunteering projects within my community. Currently focusing on Explainable AI research.

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Knoxville, TN, USA

in xie-jiaxu

SKILLS

Communication	Project Management	Finance Reporting	Snowflake Cloud	Robotics / ROS
Analytical	Develop others / mentoring	Data Engineering / SQL /	Research	Git / Linux
Agile Methodologies			AI / ML / Python	Matlab / LabVIEW

EDUCATION

M.Sc. Computer & Electrical Engineering

Marquette University 2019–2021 Milwaukee, USA

- GPA: 3.75 / 4.0
- Thesis: *Stochastic Modeling for Mobile Manipulators*
- Focus areas: Machine Learning, Deep Learning, Image Processing, Probability Theory, Algorithms

B.Sc. Electronic Engineering

Federal University of Technology – UTFPR 2014–2018 Curitiba, Brazil

- GPA: 0.81 / 1.0 (Brazilian scale)
- Thesis: image processing tool to identify insects from photos, enabling Federal Police to track drug shipments
- Built a Pac-Man game in C and an Android diet-control app in Java

High School with degree in Accounting Technician

Federal Institute of Paraná – IFPR 2011–2013 Curitiba, Brazil

PROFESSIONAL EXPERIENCE

Independent Researcher – Explainable AI

Self-directed Apr 2026 – present Knoxville, USA

Keywords: *Explainable AI, Personal Projects, AI-assisted Education*

- Independent research on Explainable AI; Development of AI-assisted tools for public education.

Data Engineer

ExxonMobil Feb 2023 – Mar 2026 Curitiba, Brazil

Keywords: *Data Pipelines, Snowflake, dbt, Cloud Data, Data Governance, STEM Outreach*

- Led MVPs and initiatives that optimized data management, contributing to multi-million dollar savings for the corporation.
- Built source-to-target data pipelines.
- Mentored junior developers.
- Managed cloud data objects, controlled access permissions, and curated data for specific use cases.
- Member of the local STEM committee: designed road maps and delivered training sessions to support women in data and programming.
- Partnership R&D project with UFRJ (Federal University of Rio de Janeiro) : exploring ML-based methods to solve data reconciliation challenges within big corporations.

R&D Engineer

Lactec ¹ May 2022 – Feb 2023 Curitiba, Brazil

Keywords: *Computer Vision, Robotics, ML, ROS, Stakeholder Management*

- Conducted R&D projects in AI, computer vision, and robotic automation.
- Delivered a high-value integrated robotic system for electric substations; wrote and secured a Phase II approval from sponsors and stakeholders.
- Built ML models and explored evaluation metrics tailored to each project scenario, which resulted in a published paper.
- Delivered internal training sessions on Computer Vision and ROS; supervised interns and conducted weekly stakeholder updates.
- Wrote and submitted papers and project proposals, including financial planning and deliverable timelines.

¹Lactec is a prestigious private research center in Brazil

Data Science / Data Engineering Mentor

📅 Harve 📅 Apr 2022 – Nov 2022 📍 Remote/Curitiba

Keywords: *Python, ML, SQL, Data Engineering, Teaching*

- Mentored students on data science projects.
- Presented technical sessions that covered Python, ML best practices, SQL, and data engineering.

Research Assistant

📅 Marquette University – COVISS Lab 📅 Apr 2019 – Dec 2021 📍 Milwaukee, USA

Keywords: *Kalman Filters, ROS, Gazebo, Matlab, NIST-sponsored Research*

- Designed and evaluated mathematical models in Matlab; built computational scenarios for robotic experiments (NIST-sponsored project).
- Implemented and integrated Kalman filter-based tracking algorithms for mobile manipulators with ROS and Gazebo.
- Wrote technical reports, conducted peer reviews, and provided weekly research updates.

Technical Assistant – R&D

📅 UTFPR & COPEL 📅 Dec 2017 – Jan 2019 📍 Curitiba, Brazil

Keywords: *LabVIEW, National Instrument devices, Photovoltaics*

- R&D project sponsored by Copel, major Brazilian electric utility company.
- Developed LabVIEW VIs for acquisition and monitoring of photovoltaic panel data (voltage, current, power) using myRIO and compactRIO, both devices from National Instruments.

EARLIER UNDERGRADUATE POSITIONS

Omatic Engenharia – Intern

Aug – Oct 2018: *Smart Cities engineering consulting*

Lactec – Electronic Engineer Intern

Oct 2016 – Jan 2018: *image processing, sensor integration, PCB manufacturing*

UTFPR – IT Intern (DEINFRA department)

Sep 2015 – Mar 2016: *software and hardware support*

PARALLEL ACTIVITIES & RESEARCH

Freelance projects

Designed web interfaces using Javascript and Angular; built datasets for NLP project using string matching and Python libraries.

PICME Program

Scientific research in Optimal Control, Aug 2014 – Jul 2016

Volunteer TA – Calculus II (UTFPR)

Apr – Dec 2015: student support during office hours

Volunteer Teacher – Maths & Physics

Formação Solidária, Apr – Aug 2015: support to public high school students.

Fundação Arauária – Junior Researcher

Binary Gas Reactive Study at IFPR, 2011–2012.

PIC – Scientific Initiation

Brazilian Math Olympiad for Public Schools program, concluded in 2014.

SCIENTIFIC PUBLICATIONS

- Co-authored – *Artificial Intelligence-Assisted Methodology for Dataset Reduction Applied to the Establishment of Power Interruption Limits in Brazil*¹.
- Master thesis – *Stochastic Modeling for Mobile Manipulator*².
- Co-authored – *A Monitoring and Management System for Medium-Scale Photovoltaic Plants*³.
- Conference paper – *Pattern recognition in botanical micro-traces for forensics applications using OpenCV*⁴.
- Conference paper – *Optimal control of water flow in an ice tray*⁵.

AWARDS

OBMEP – Bronze Medal

Brazilian Math Olympics for Public Schools, 2008 & 2013

OBMEP – Honorable Mention

Brazilian Math Olympics for Public Schools, 2012

Accounting Intern

Auditop, 2012: financial transaction documentation and account analysis

LANGUAGES

Portuguese

Native

English

Fluent

Cantonese

Native

Spanish

Basic

Works Cited

- [1] • Rhafael Freitas da Costa, Gabriela Rosalee Weigert-Dalagnol, Débora Cintia Marcilio, Lúcio de Medeiros, Eulerson José da Sila Junior, **Xie Jiayu**, et al. "Artificial Intelligence-Assisted Methodology for Dataset Reduction Applied to the Establishment of Power Interruption Limits in Brazil". In: *Energies* 16.19 (2023). ISSN: 1996-1073. DOI: 10.3390/en16197012. URL: <https://www.mdpi.com/1996-1073/16/19/7012>.
- [2] • Xie Jiayu. "Stochastic Modeling for Mobile Manipulators". Advisor: Dr. Henry Medeiros. MA thesis. Marquette University, 2021. URL: https://epublications.marquette.edu/theses_open/687/.
- [3] • Clayton H. da Costa, Marcelo P. Rodrigues, Guilherme D. Yamada, Gabriel V. Rodrigues, **Xie Jiayu**, Guilherme L. Moritz, et al. "A Monitoring and Management System for Medium-Scale Photovoltaic Plants". In: *2019 IEEE PES Innovative Smart Grid Technologies Conference - Latin America (ISGT Latin America)*. 2019, pp. 1–6. DOI: 10.1109/ISGT-LA.2019.8895463.
- [4] • Xie Jiayu and Rubens A. De Faria. "Reconhecimento de padrões de imagens em microvestígios para aplicações forenses utilizando a ferramenta OpenCV". in: *XXIV Congresso Nacional de Criminalística, VII Congresso Internacional de Perícia Criminal e XXIV Exposição de Tecnologias Aplicadas à Criminalística*. 2017.
- [5] • Xie Jiayu and Joao L. Gonçalves. "Controle ótimo do fluxo de água em uma forma de gelo". In: *XXXVI Congresso Nacional de Matemática Aplicada e Computacional – CNMAC 2016*. 2016.